MIL-S-24251B
17 September 1979
SUPERSEDING
MIL-S-24251A
1 July 1968

MILITARY SPECIFICATION

SHIELDS, RETAINERS (BASES), AND ADAPTERS, ELECTRON TUBE, HEAT DISSIPATING

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for heat dissipating electron tube shields and associated retainer (bases), and adapters for use in electronic equipment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/2	-	Aluminimum Alloy, 3003, Plate and Sheet.
QQ-A-250/8	-	Aluminimum Alloy, 5052, Plate and Sheet.
QQ-A-591	-	Aluminimum Alloy Die Castings.
QQ-B-613	-	Brass, Leaded and Non-Leaded, Plate, Rolled Bar, Sheet and Strip.
QQ-C-530	-	Copper-Beryllium Alloy Bar, Rod, and Wire (Copper Alloy Number 172).
QQ-C-533	-	Copper-Beryllium Alloy Strip (Copper Alloy Numbers 170 and 172).
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-S-781	-	Strapping, Steel and Seals.
QQ-W-423	-	Wire, Steel, Corrosion-Resisting.
WW-T-700/6	-	Tube, Aluminum Alloy, Drawn, Seamless, 6061.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPB-B-585	-	Box, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-B-676	-	Boxes, Setup.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-E-1	-	Electron Tubes, General Specification for.
MIL-P-116	-	Preservation, Methods of.
MIL-C-5541	-	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-12883	-	Sockets, for Plug-In Electronic Components and Accessories, General Specification for.
MIL-C-13924	-	Coating, Oxide, Black, for Ferrous Metals.
MIL-I-17214	-	Indicator, Permeability, Low-Mu (Go-No-Go).
MIL-C-45662	-	Calibration System Requirements.

(See supplement 1 for list of associated detail specifications, specification sheets).

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-794	-	Part and Equipment Procedures for Packaging and Packing of.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.

COMMERCIAL

EIA-STD RS-209	-	EIA Standards for Electron Tubes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Specification sheets. The individual part requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the individual specification sheet, the latter shall govern.

3.2 Material. The material shall be as specified (see 3.1). When a definite material is not specified, a material shall be used which will enable the tube shields, retainers (bases), and adapters to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Dissimilar metals. Where dissimilar metals are used in intimate contact with each other, protection against corrosion shall be provided. Dissimilar metals shall be as defined in MIL-STD-454, requirement 16.

3.3 Design, construction and physical dimensions. The tube shields, retainers (bases), and adapters shall be of the design, construction, and physical dimensions specified (see 3.1). The design shields and retainers (bases) and adapters shall be such that they will mount to the chassis using the same hardware (including screws), as that of the applicable tube socket.

3.3.1 Finish. Unless otherwise specified (see 3.1), tube shields, retainers (bases), and adapters shall be treated with a black matte finish. That portion of the mating shield and retainer (base) or retainer (base) and chassis where low electrical resistance is required may be finished with other than black matte.

3.3.2 Liners. Shields shall be constructed in such a way that the liners are securely retained in position within the shield (see 3.5.3). Liners with circumferential corrugations or other configurations which could cause tube breakage and that do not permit spring action within the shield to allow for maximum tube diameter variations, are not acceptable.

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3.3.3 Installation and removal. Design of the shield shall be such that installation and removal of the shield from a seated electron tube shall not scratch or damage the electron tube envelope when tested as specified in 4.5.3.4. Unless otherwise specified, no tool shall be required for installation or removal of the shield from the tube or retainer (base) or adapter.

3.3.4 Spring retention. When used (see 3.1), the coil spring shall be securely retained within the shield. The free end of the spring, or the loop that makes contact with the electron tube envelope shall be at least seven-eights of a complete turn and the spring tip shall not touch the tube envelope.

3.3.5 Retainer (bases) and adapter. The retainer (base) and adapter design shall be such that it rigidly retains the tube shield during all tests and examinations of this specification.

3.4 Permeability. All parts shall be made from materials which are classed as nonmagnetic. The permeability of the tube shield assembly shall be less than 2 (air = 1.0) (see 4.5.2).

3.5 Installation, locking and unlocking torque or load, and removal force.

3.5.1 Initial force. The installation, locking and unlocking torque or load, and removal force necessary to install and remove the shield over the applicable slug (see figure 1) and retainer (base), and adapter shall be not greater than specified on the applicable specification sheet (see 3.1), when tested in accordance with 4.5.3.1.

3.5.2 Force at temperature extremes. The installation, locking and unlocking torque or load, and removal force shall be not greater than 150 percent nor less than 50 percent of the initial force when tested at 85° ±5°C and -55° ±5°C, in accordance with 4.5.3.2.

3.5.3 Fatigue. When tested in accordance with 4.5.3.3, the installation or removal force shall be not greater than 125 percent nor less than 50 percent of the initial force and the liner shall remain in position within the shield.

3.6 Vibration. When tested in accordance with 4.5.4, the shield shall not loosen from the retainer (base) or adapter. There shall be no evidence of damage to the shield, or retainer (base) and adapter, nor the glass envelope of the electron tube, during installation, during the test, or upon removal after the test.

3.7 Shock. When tested in accordance with 4.5.5, the shield shall not loosen from the retainer (base) or adapter as applicable, and there shall be no evidence of damage to the shield, tube socket, shield retainer (base), or the glass envelope of the electron tube.

3.8 Heat resistance. When shields, retainers (base) and adapters are tested in accordance with 4.5.6, there shall be not peeling, blistering, chipping, or pitting of the finish.

3.9 Heat dissipation. The average reduction of a minimum of two surface temperature readings of the instrumented tube shall be not less than the value specified (see 3.1), when tested in accordance with 4.5.7.1.

3.10 Salt spray (corrosion). When shields, retainers (base) and adapters are tested as specified in 4.5.8, there shall be no evidence of corrosion and the finish shall not peel, blister, chip, or pit.

3.11 Electrical contact. The dc resistance from the shield to the chassis shall be not greater than 1 ohm when tested in accordance with 4.5.9.

3.12 Marking. Each tube shield, retainer (base), or adapter shall be permanently and legibly marked in accordance with MIL-STD-130 with the following:

a. Military part number.
b. Date and source code in accordance with MIL-STD-1285.

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3.13 Workmanship. Shields and associated parts shall be processed in such a manner, as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, chips, spalls and other defects that will affect life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The contractor shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspection and testing of electron tube shields, retainers (bases) and adapters shall be classified as follows:

a. Quality conformance inspection (see 4.4).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be made in accordance with the general requirements of MIL-STD-202.

4.4 Quality conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection.

4.4.1.1 Inspection lot. An inspection lot shall consist of tube shields, retainers (bases), or adapters of the same design and size, produced under essentially the same conditions and offered for inspection at one time.

4.4.1.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.1.3 Group A inspection. Group A inspection shall consist of the inspections specified in table I, in the order shown.

TABLE I. Group A inspection.

Examination	Requirements paragraph	Method paragraph	AQL (percent defective) Major	Minor
Visual and mechanical examination - - - - -	3.2 thru 3.3.5 and 3.13	4.5.1	1.0	4.0

4.4.1.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table I. Major and minor defects shall be as defined in MIL-STD-105.

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4.4.1.4 Group B inspection. Group B inspection shall consist of the inspections specified in table II, in the order shown, and the sample shall be selected from inspection lots that have passed group A inspection.

TABLE II. Group B inspection.

Test	Requirement paragraph	Method paragraph
Heat resistance	3.8	4.5.6
Heat dissipation	3.9	4.5.7
Permeability	3.4	4.5.2
Installation and removal	3.4.3.3	4.5.3.4
Salt spray (corrosion)	3.10	4.5.8
Electrical contact	3.11	4.5.9
Installation, locking and unlocking torque or load, and removal force	3.5	4.5.3
Initial force at 25°C	3.5.1	4.5.3.1
Force at temperature extremes	3.5.2	4.5.3.2
Fatigue	3.5.3	4.5.3.3
Vibration	3.6	4.5.4
Shock	3.7	4.5.5

4.4.1.4.1 Sampling plan. Sample units shall be selected from tube shields, retainers, or adapters that have passed group A inspection. Two sample units shall be selected and tested every 24 months. The period shall start the date the last test was satisfactorily completed on each of the individual units and run for 24 months.

4.4.1.4.2 Test routine. Group B inspection shall be performed at the start of production, and each 24-month period thereafter. If production of a particular shield, retainer, or adapter has been suspended for 24 months or more, group B inspection shall be again required at the time production is started. In such a case, inspection shall be performed on sample units selected from the first lot of the new production presented for acceptance, and after each subsequent 24-month period.

4.4.1.4.3 Acceptance criteria. No failures shall be permitted.

4.4.1.4.4 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on contract or order.

4.4.2 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

4.5 Method of examination and tests.

4.5.1 Visual and mechanical examination. Shields and associated parts shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements specified herein (see 3.2, 3.3, 3.12 and 3.13).

4.5.2 Permeability. The permeability of the tube shield assembly shall be measured with an indicator conforming to MIL-I-17214 to determine conformance with 3.4.

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4.5.3 Installation, locking and unlocking torque, or load and removal force.

4.5.3.1 Initial force at 25°C. The force necessary to place, lock, unlock, and remove the shield over the applicable slug and retainer (base) or adapter shall be measured at 25° ±5°C (see figure 1 and 3.5.1).

4.5.3.2 Force at temperature extremes (see 3.5.2). The slug shown on figure 1, the shield and retainer base or adapter shall be conditioned at 85° ±5°C for 2 hours after which the test of 4.5.3.1 (at this temperature extreme) shall be performed within 1 minute after opening the chamber. After completion of the 85°C test, the slug, shield, and retainer (base) or adapter shall then be conditioned at -55° ±5°C for 2 hours after which the test specified in 4.5.3.1 (at this temperature extreme) shall be repeated within 1 minute after opening the chamber.

4.5.3.3 Fatigue test. After the shield is properly installed and removed from the slug and applicable retainer (base) or adapter 100 times, the tests specified in 4.5.3.1 shall again be performed (see figure 1 and 3.5.3).

4.5.3.4 Installation and removal. The shield with all associated parts shall be installed and removed 25 times over a mounted tube of proper size (see table IV). The electron tubes, liner, and coil springs (when used) shall be examined after the test for compliance with 3.3.2, 3.3.3 and 3.3.4.

4.5.4 Vibration (see 3.6). The shields and the retainer (base) or adapter (when applicable) shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

a. Mounting method - Normal mounting (with a tube in the socket).
b. Test condition - B.
c. Electrical load - None.
d. Measurements during and after vibration - None.

After the test, shields, retainers (bases) or adapters and electron tube glass envelopes shall be examined for evidence of damage or loosening from the base.

4.5.5 Shock (see 3.7). The shields and the retainer (base) or adapter (when used) shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

a. Mounting method - Normal mounting (with a tube in the socket).
b. Test condition - A.
c. Electrical load - None.
d. Measurements after shock - None.

After the test, the shields, retainers (bases) or adapters and electron tube glass envelopes shall be examined for evidence of damage or loosening from the base.

4.5.6 Heat resistance (see 3.8). The shields, retainers (base) or adapters shall be subjected to an ambient temperature of 200° ±5°C for 48 hours. The shields, retainers (base) or adapters shall then be conditioned for 1 hour at room ambient temperature (see 4.3). At the end of this 1-hour interval, the shields and associated parts shall be examined for evidence of damage to the body finish (peeling, blistering, chipping or pitting).

4.5.7 Heat dissipation (see 3.9). An instrumented glass tube (see appendix A and figure 3) shall be used. The glass material and dimensions shall be identical to those of the envelope size for which the tube shield is designed (see table IV). The instrumented tube shall be mounted vertically in the appropriate MIL-S-12883 socket, which is mounted with the proper retainer (base) or adapter (when applicable), on a heat sink which is maintained at a temperature of 25° ±5°C.

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4.5.7.1 Envelope stabilization. These measurements shall be made in an ambient temperature of 25° ±5°C, and with no draft on the test setup. Temperature shall be measured with a bridge null detecting device with the tube mounted in the socket on a constant temperature (25° ±5°C) heat sink under a 1x1x1 ft plastic cover. The instrumented tube, (see Appendix A) shall be wired into the circuit shown on figure 2 with the heater voltage adjusted to the required nominal voltage. Plate voltage shall be as shown in table III. The grid bias shall be adjusted to increase the plate current until both the total dissipation watts (within the range specified in table III) and the bare bulb temperature (within the tolerance specified in table III), are reached. The stabilized bulb temperature as indicated by the bulb thermocouple shall be recorded. The shield assembly shall be mounted on its base. The stabilized bulb temperature with the shield mounted shall again be recorded. The shield shall be rotated to at least two other different positions, except those shields which have only two rotatable positions, which shall be only one other position. Measurements shall be repeated for each increment of rotation. During the test, the heat sink and ambient temperatures for each separate shield test shall be recorded.

TABLE III. Instrumented tube parameters.

Envelope size 1/	Tube type	Test circuit	Plate voltage Ep-Dc volts maximum	Plate current Dc milliamps (max/plate)	Total dissipation (filament and plate) Nominal-Watts ± 10%	Bare bulb temperature ° ±5°C
T2x3 (Flat bulb) - - - -	6029	7	135	---	1.14	100
T3 (Round bulb, 8 pin) -	5902	7	150	---	6.19	220
T3 (Flat press, 7 pin-in-line) - - - - - - - -	5702WB	7	150	---	2.67	145
T5-1/2 Short - - - - - -	5654	1	130	---	3.1	115
T5-1/2 Medium - - - - -	6AN5WA	1	130	---	9.1	215
T5-1/2 Long - - - - - -	6005	2	250	---	12.9	240
T6-1/2 Short - - - - - -	5670	3	250	---	4.8	130
T6-1/2 Medium - - - - -	5687WA	4	200	---	13.2	225
T6-1/2 Long- - - - - - -	6216	5	200	---	18	230
T6-1/2 Ex-Long - - - -	7189A	6	250	---	19.5	235
T9- - - - - - - - - - - -	5Y3WGTA	8	2/	290	23.5	225
T11 - - - - - - - - - - -	6L6WGB	7	300	---	33.1	250
T12 - - - - - - - - - - -	5U4GB	8	2/	700	32.7	200
T14 - - - - - - - - - - -	5R4WGB	8	2/	490	45.5	250
T16 - - - - - - - - - - -	6336A	9	300	---	54.3	250

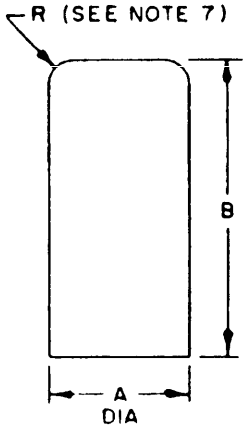
1/ Conforming EIA STD RS-209.

2/ Current limitation.

4.5.8 Salt spray (corrosion). Shields and associated parts shall be tested in accordance with method 101, test condition B, of MIL-STD-202. Upon completion of the test, the shields and associated parts shall be examined for evidence of corrosion or other defects as specified in 3.10.

4.5.9 Electrical contact. The shield retainer (base), adapter and socket shall be mounted to a bare metal chassis in the normal manner. An appropriate tube shall be mounted in the assembly. The protective coating may be removed from a point on the upper part of the shield so that good electrical contact may be made with the ohmmeter probe. The resistance shall be measured between shield and chassis using an ohmmeter having an accuracy of ±3 percent of indication or better (see 3.11).

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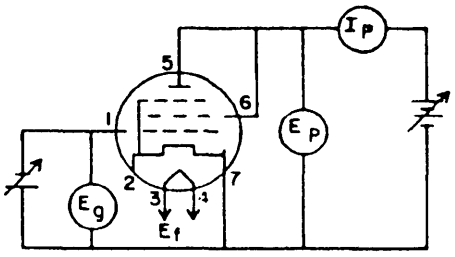


Tube outline	Inches		
	A ±.001(.03)	B ±.010(.25)	R ±.031(.79)
T3	.380 (9.65)	1.100 (27.94) 1.400 (35.56) 1.625 (41.28)	.080 (2.03)
T5½	.730(18.54)	1.125 (28.58) 1.500 (38.10) 2.000 (50.80)	.156 (3.96)
T6½	.840(21.34)	1.125 (28.58) 1.562 (39.67) 2.000 (50.80) 2.437 (61.90)	.281 (7.14)
T9	1.150(29.21)	1.750 (44.45) 2.062 (52.37) 2.312 (58.72) 2.500 (63.50) 2.688 (68.28) 2.750 (69.85) 3.062 (77.77) 3.250 (82.55)	.400(10.16)
T11	1.410(35.81)	2.937 (74.60) 3.312 (84.12) 3.687 (93.65) 4.062(103.17)	.470(11.94)
T12	1.560(39.62)	2.375 (60.33) 2.500 (63.50) 2.625 (66.68) 2.750 (69.85) 2.938 (74.63) 3.000 (76.20) 3.125 (79.38) 3.250 (82.55) 3.312 (84.12) 3.500 (88.90) 3.688 (93.68) 3.750 (95.25) 3.875 (98.43)	.500(12.70)
T14	1.830(46.48)	4.125(104.78) 4.813(122.20)	.590(14.99)
T16	2.050(52.07)	4.175(106.05) 4.750(120.65)	.650(16.51)

NOTES:
1. Dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Material: Tool steel, hardened to 56-61C.
5. Finish: Hard chrome plate.
6. Mounting method optional (screw and pins).
7. Radius shall be blended smoothly into A diameter.

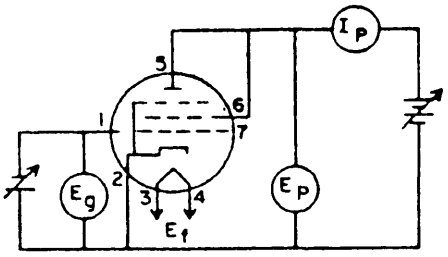
FIGURE 1. Test slugs for installation and removal force tests.

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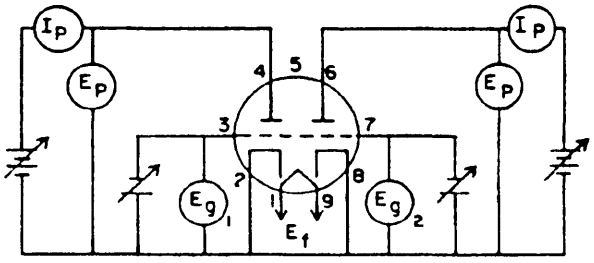
E_f = 6.3 VAC or VDC
E_g = 0-35 VDC

Circuit 1 - For tube type 5654 and 6AN5WA.



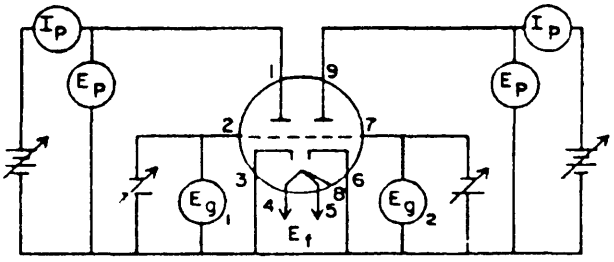
E_f = 6.3 VAC or VDC
E_g = 0-35 VDC

Circuit 2 - For tube type 6005



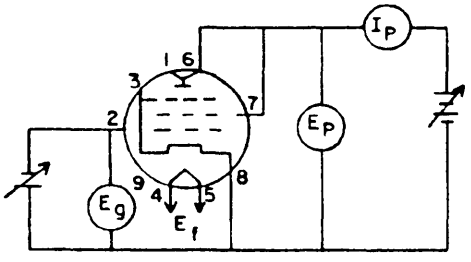
E_f = 6.3 VAC or VDC
E_g , E_{g2} = 0-35 VDC

Circuit 3 - For tube type 5670



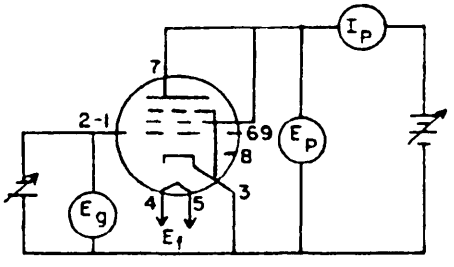
E_f = 6.3 VAC or VDC
E_{g1}, E_{g2} = 0-35 VDC

Circuit 4 - For tube type 5687WA.



E_f = 6.3 VAC or VDC
E_g = 0-35 VDC

Circuit 5 - For tube type 6216



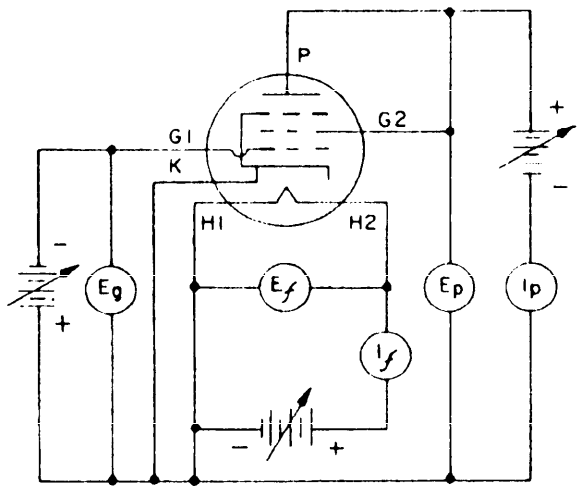
E_f = 6.3 VAC or VDC
E_g = 0-35 VDC

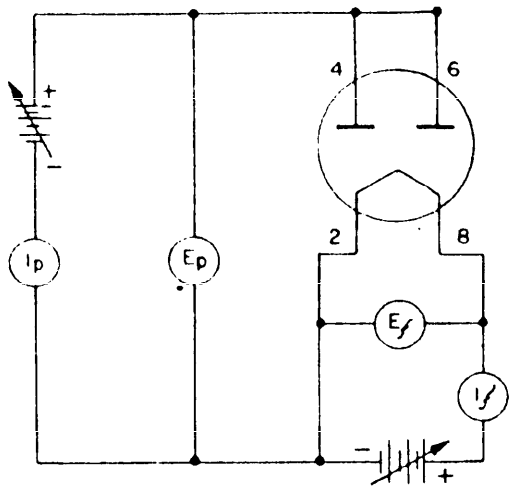
Circuit 6 - For tube type 7189A.

FIGURE 2. Test circuits.

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3.3.18.2 Specification Manufacturer's requirements. The Government





Ef = 5.0±.5 VDC

Circuit 8 – For tube types 5Y3WGTA, 5U4GB and 5R4WGB.

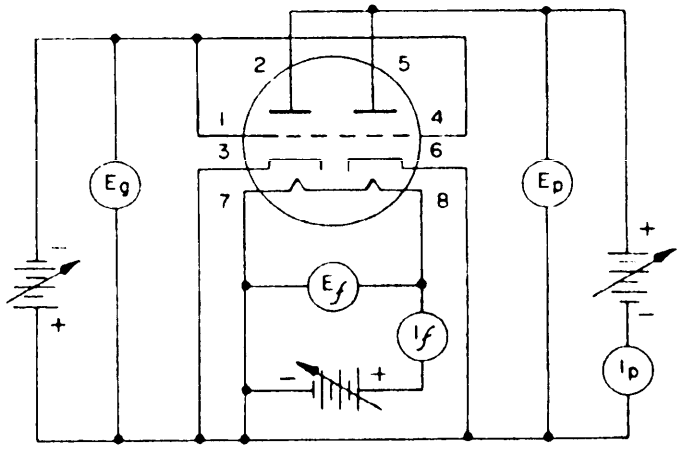
	Element								
Tube	H1	H2	K	G1	G2	G3	P	——	Ef(VDC)
6L6WGB	2	7	8	5	4	I.C*	3	Tube	6.3±.6
6029	2(–)	4(+)	–	3	–	–	1	pin	1.25±.05
5902	3	6	2	1	7	I.C*	5	No.	6.3±.3
5702WB	3	4	6	7	2	5	1		6.3±.6

Eg = 6L6WGB(–20VDC); 6029(–20VDC); 5902(–25VDC); 5702WD(–15 VDC)

NOTES:
1. Observe polarity of 6029 filament connections: H1(–); H2(+).
2. (*) I.C. means element internally connected.

Circuit 7 – For tube types 6L6WGB, 6029, 5902, and 5702WB.

Grid voltage shall be initially set at maximum value to avoid damage to the tube. After reaching maximum plate voltage, slowly decrease grid voltage until either maximum total power, plate current or bulb temperature is obtained. Care shall be taken to prevent the tube from oscillating. It may be necessary to shield the plate and grid lead. If the tube oscillates, erroneous readings may be obtained. Both halves of dual triode tubes shall be set at equal power levels.



Ef = 6.3±.6 VDC
Eg = –200 VDC

Circuit 9 – For tube type 6336A

FIGURE 2. Test circuits - Continued.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A, C of MIL-P-116, or commercial of MIL-STD-1188, or as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Shields, retainers, and adapters shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Shields, retainers, and adapters shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packs. Each shield, retainer, and adapter shall be individually unit packed in accordance with MIL-P-116, method III insuring compliance with the applicable requirements of that specification. The unit container for items exceeding one inch in smallest dimension shall conform to PPP-B-566 or PPP-B-676.

5.1.1.5 Intermediate packs. Except as specified herein, shields, retainers, and adapters, packaged in accordance with 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities, shall be of minimum tare and cube, and shall contain multiples of five unit packs, not to exceed 100 unit packs. No intermediate packs are required when the total quantity shipped to a single destination is less than 100 unit packs or when boxes are used as the unit containers.

5.1.2 Level C. The level C preservation for shields, retainers, and adapters shall conform to the MIL-STD-794 requirements for this level.

5.1.3 Commercial. The commercial preservation of shields, retainers, and adapters shall be in accordance with the requirements of MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, C of MIL-P-116, or commercial of MIL-STD-1188, or as specified (see 6.2).

5.2.1 Level A. The packaged shields, retainers, and adapters shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, class 1, or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged shields, retainers, and adapters shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The level C packing for shields, retainers, and adapters shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Commercial. The preserved shields, retainers, and adapters shall be packed in accordance with the requirements of MIL-STD-1188.

5.3 Marking.

5.3.1 Levels A, B, and C. In addition to any special marking required by the contract (see 6.2), each unit pack and intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of MIL-STD-1188.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

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5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.2.

5.4.3 Army procurements.

5.4.3.1 Level A unit and intermediate packs. All unit and intermediate containers shall be either weather (or water) resistant or overwrapped with waterproof barrier materials (see 5.1.1.4 and 5.1.1.5).

5.4.3.2 Level A and Level B packing. For level A packing the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A and the containers shall be banded (see 5.2.1 and 5.2.2).

6. NOTES

6.1 Intended use. Heat dissipating electron tube shields are used primarily for lowering the envelope operating temperature, but in addition, may be used to provide electrostatic shielding and retention under shock and vibration.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of the applicable specification sheet.
c. Complete military part number.
d. Inspection of commercial packaging (see 4.4.2).
e. Levels of preservation and packing required (see 5.1 and 5.2).
f. Special marking, of required (see 5.3.1).

6.3 Cross index. Tube shields, retainers (bases), and adapters contained herein are cross indexed to those formerly covered by MIL-S-19786, MIL-S-9372 and applicable Military Standards, as shown in table IV.

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TABLE IV Cross index of superseded types.

Shield	MIL-S-24251/ Retainer (base)	MIL-S-19786	MIL-S-9372
1-1		S0761AV01	
1-2	2-1	S0762AV01	
1-3		S0765AV01	
1-4		S0966AV02	
1-5	2-2	S0967AV02	
1-6		S0968AV02	
3-1		S0761AV01	
3-2		S0762BV01	
3-3		S0765BV01	
3-4		S0966AV02	
3-5		S0967BV02	
3-6		S0968BV02	
3-7			MS24230
	4-1		MS24231-1
	4-2		MS24231-2
5-1			MS24232-1
5-2			MS24232-2
5-3			MS24232-3
5-4			MS24232-4
5-5			MS24232-5
5-6			MS24232-6
6-1		S0761CV0D and S0761XV00	MS24233-1
6-2		S0762CV00 and S0762XV00	MS24233-2
6-3		S0765CV00 and S0765XV00	MS24233-3
6-4		S0966CV00 and S0966XV00	MS24233-4
6-5		S0967CV00 and S0966XV00	MS24233-5
6-6		S0968CV00 and S0968XV00	MS24233-6
6-7			MS24233-7
7-1			MS24270-1
7-2			MS24270-2
7-3			MS24270-3
7-4			MS24270-4
7-5			MS24270-5
7-6			MS24270-6
7-7			MS24270-7
7-8			MS24270-8
	8-1		MS24271-1
	8-2		MS24271-2
9-1			MS24272-1
9-2			MS24272-2
9-3			MS24272-3
9-4			MS24272-4
	10-1		MS24273-1
11-1			MS24274-1
11-2			MS24274-2
11-3			MS24274-3
11-4			MS24274-4
11-5			MS24274-5
11-6			MS24274-6
11-7			MS24274-7
11-8			MS24274-8
12-1			MS24275-1
12-2			MS24275-2
12-3			MS24275-3
12-4			MS24275-4

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TABLE IV. Cross index of superseded types. -Continued

Shield	MIL-S-24251/ Retainer (base)	MIL-S-19786	MIL-S-9372
12-5			MS24275-5
12-6			MS24275-6
12-7			MS24275-7
12-8			MS24275-8
12-9			MS24275-9
13-1			MS24276-1
13-2			MS24276-2
13-3			MS24276-3
13-4			MS24276-4
13-5			MS24276-5
13-6			MS24276-6
13-7			MS24276-7
13-8			MS24276-8
	14-1		MS24277-1
	14-2		MS24277-2
15-1			MS24278-1
16-1			MS24279-1
	17-1		MS24280-1

6.4 Changes from previous issue. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - CG, MC, AS

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3203)

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APPENDIX A

INSTRUMENTED TUBES

10. SCOPE

10.1 This appendix covers the instrumented tubes for use in the heat dissipation tests specified in 4.5.7. The tube type recommended to represent each bulb type was chosen for its high power dissipation. The tube types are listed in table III

10.2 Tube thermocouple instrumentation.

10.2.1 Locating hot spots. A temperature sensitive paint shall be applied to the tube envelope. The appropriate value paint shall be determined by trial and error to provide an accurate indication of the hot spot location when the tube is operated at or near its maximum power dissipation rating and the hot spot is located by visual observation.

10.2.2 Preparing the bulb. The tube envelope diameter shall be measured with a micrometer and recorded. The tube shall be prepared for grooving by masking the entire bulb with masking tape, leaving only a 1/16 inch ± 1/64 inch strip uncovered, running lengthwise along the bulb, and through the hot spot. The upper end of the bare strip shall stop 1/8-inch to 3/16-inch from the beginning of the curved shoulder at the top of the tube. The bare strip shall be allowed to run out at the bottom of the tube.

10.2.3 Etching the groove. A sand blaster with a 1/4 to 1/2-inch-diameter nozzle may be used. The etching material may be No. 120 or 220 grit aluminum oxide powder. Air pressure may be 45 psi. The nozzle may be positioned 4-inches from the tube surface, so that the etching powder will impinge at right angles to the surface to be etched.

10.2.3.1 Approximately 25 to 50 passes may be required with the nozzle over the exposed glass. Each pass shall be extended beyond both ends of the tube envelope to prevent overetching at the ends of the groove. On the 6336A tube, it may be necessary to roughen (by sandblasting) a small surface area, delta in shape, at the lower end of the groove on the outside of the glass, to serve as a better anchor for the thermocouple and protective tube.

10.2.3.2 The depth of the groove shall be 0.010-inch + .003-inch, -.001-inch. Periodic measurements of groove depth shall be made, as the speed of etching will vary with the cleanliness, age, and moisture content of the etching material.

10.2.4 Thermocouple installation. The thermocouple used shall be a welded junction of 36 gage iron and constantan wire. The thermocouple shall be laid into the groove with its junction on the hot spot of the tube, held down with a spring or other means and the leads running out of the lower end of the groove. The groove shall be filled with Sauerisen, Insa-lute #1 cement, or equivalent, and the cement allowed to dry at 100° C for 24 hours or 36 to 48 hours at room temperature. The excess shall then be scraped off to restore the original contour of the tube envelope.

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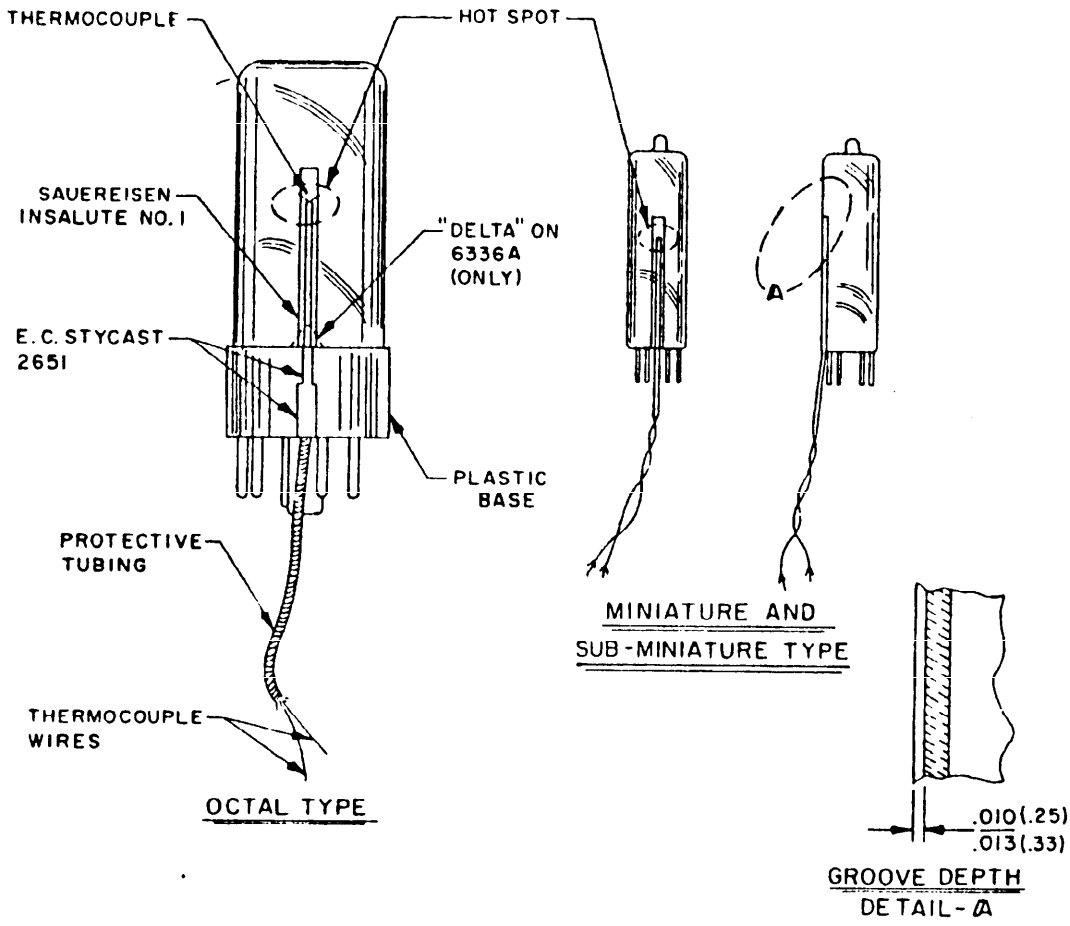


FIGURE 3. Instrumented tubes.

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